

## SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS  
COURSE CODE: CSA07

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### COURSE OUTCOME COVERED

**CO2:** Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

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Code of Conduct:

I B. Chaitanya Prakash Reddy (A2425255) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

Chaitu  
Signature of the student:

| Criteria                  | Weightage | Q1 10marks | Q2 10marks | Q3 10marks | Q4 10marks |
|---------------------------|-----------|------------|------------|------------|------------|
| Problem Understanding     | 20%       |            |            |            |            |
| Method Selection          | 20%       |            |            |            |            |
| Solution Process          | 25%       |            |            |            |            |
| Accuracy of Calculation   | 20%       |            |            |            |            |
| Interpretation of Results | 15%       |            |            |            |            |

Total Marks 29 /40

[Signature]  
Signature of the Faculty

Q1.

Given :

- IP Address Block : 192.168.20.0/24
- Lab A : /26
- Lab B : /27
- Lab C = /28.

### Lab A (/26)

- Network Address : 192.168.20.0
- First Host : 192.168.20.1
- Last Host : 192.168.20.62
- Broadcast Address : 192.168.20.63
- Total IP Address : 64
- usable Host Address : 62.

### Lab B (/27).

Next available subnet starts at 192.168.20.96

- Network Address : 192.168.20.96
- First Host : 192.168.20.97
- Last Host : 192.168.20.110
- Broadcast Address : 192.168.20.111
- Total IP Addresses : 16
- usable Host Addresses : 14.

## LAB C (/28)

- Network Address : 192.168.20.64
- First Host : 192.168.20.65
- Last Host : 192.168.20.94
- Broadcast Address : 192.168.20.95
- Total IP Addresses : 32

Conclusion:- The network 192.168.20.0/24 is successfully divided into three fixed-size subnets. Lab A 62 hosts, Lab B 30, Lab C 14.

Q2.

Given :

- System 1 IP Address : 192.168.1.130
- System 2 IP Address : 192.168.1.190
- ~~system~~ subnet mask : 255.255.255.192 (/26).

Step 1: Subnet size

- subnet Mask = /26

- Block size = 64

- subnets are :
  - 192.168.1.0 - 192.168.1.63
  - 192.168.1.64 - 192.168.1.127
  - 192.168.1.128 - 192.168.1.191
  - 192.168.1.192 - 192.168.1.255.

System 1 (192.168.1.130)

- Network Address : 192.168.1.128
- First Host : 192.168.1.129



### Sales Department (/26)

- Network Address : 172.16.0.128
- First Host : 172.16.0.129
- Last Host : 172.16.0.190
- Broadcast : 172.16.0.191
- Usable Hosts : 62.

### IT Department (/27)

- Network Address : 172.16.0.192
- First Host : 172.16.0.193
- Last Host : 172.16.0.222
- Broadcast : 172.16.0.223
- Usable Hosts : 50.

### Admin Department (/28).

- Network Address : 172.16.0.224
- First Host : 172.16.0.225.
- Last Host : 172.16.0.238.
- Broadcast : 172.16.0.239
- Usable Hosts : 14.

conclusion:- The network 172.16.0.0/16 has been successfully subnetted for the HR, Sales, IT and Admin departments. each department receives the required subnet size with the correct network address.

Last Host : 192.168.1.190

Broadcast Address : 192.168.1.191.

∴ System 2 also belongs to the subnet 192.168.1.128/26.

System 2 (192.168.1.190).

- Network Address : 192.168.1.128

- First Host : 192.168.1.129

- Last Host : 192.168.1.190

- Broadcast : 192.168.1.191.

∴ System 2 also belongs to the subnet 192.168.1.128/26.

Q3.

Given :

- IP block : 172.16.0.0/16

- HR → /25

- Sales → /26.

- IT → /27

- Admin → /28.

Step 1 : Assign subnets.

HR Department (/25)

- Network Address : 172.16.0.0

- First Host : 172.16.0.1

- Last Host : 172.16.0.126

- Broadcast : 172.16.0.127

- Usable Host : 126.

Given:

- Network : 172.16.10.0/24
- Subnet Mask : /27 (255.255.255.204)
- Total subnets : 8
- usable host per subnet : 30

IP Address : 172.16.10.78

- subnet : 172.16.10.64 /27
- Network Address : 172.16.10.64
- Host Range : 172.16.10.65 - 172.16.10.94
- Broadcast : 172.16.10.95

check IP : 172.16.10.95 :

- 172.16.10.95 is the broadcast address ; so it does not belong to the valid host range

Conclusion: The IP 172.16.10.78 belongs to the 172.16.10.64/27 subnet, which support 30 usable hosts. The IP 172.16.10.95 is the broadcast address and cannot be assigned a host.

5Q.

Given IPv6 Address:

2001:0db8:0000:0000:ff00:0042:8329

compressed form : 2001:db8:ff00:42:8329



Expanded form:

2001 : 0db8 : 0000 : 0000 : 0000 : ff00 : 0042 : 8329

Network Prefix (/64) : 2001 : 0db8 : 0000 : 0000 :: /64

Total Host Address :  $2^{64}$  (18,446,744,073,709,551,616.)

Conclusion:

The IPv6 address is compressed as 2001 : db8 : : ff00 : 42 : 8329, its /64 network prefix is 2001 : db8 : 0000 : 0000 :: /64 and it supports  $2^{64}$  possible host addresses.